Supplementary Material for Distributional Causal Mediation via Conditional Generative Modeling

A Role of Mediator Ordering in IPSE

The IPSEs in the main text use an index-based partition around the target mediator. This partition is a bookkeeping device for defining ordered IPSEs and does not require the mediators to follow a complete causal ordering. However, different mediator indexings may define different mediator-specific estimands. When a scientifically meaningful mediator ordering is available, the ordering can be chosen to reflect that structure. When no ordering is clearly preferred, one may report IPSEs across a small set of plausible orderings or use the order-averaged summary defined below.

For a pre-specified set Π of mediator orderings, define

$$\overline{\mathrm{IPSE}}_j^{\Psi} = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathrm{IPSE}_{\pi,j}^{\Psi},$$

where π denotes a permutation of the mediator indices and $\mathrm{IPSE}_{\pi,j}^{\Psi}$ is the ordered IPSE for mediator j under that permutation. This average depends on the chosen set Π and is used only as a descriptive sensitivity summary. In the main text, we use the given mediator indexing as a reference ordering to present the IPSE construction.

For example, consider two mediators indexed as (M_1, M_2). Under this ordering, the two mediator-specific effects are

$$\begin{aligned} \mathrm{IPSE}_1^{\Psi} &= \Psi\Big(P_{Y_{1\tilde{M}_{11}\tilde{M}_{21}}}, P_{Y_{1\tilde{M}_{10}\tilde{M}_{21}}}\Big), \\ \mathrm{IPSE}_2^{\Psi} &= \Psi\Big(P_{Y_{1\tilde{M}_{10}\tilde{M}_{21}}}, P_{Y_{1\tilde{M}_{10}\tilde{M}_{20}}}\Big). \end{aligned}$$

Thus, each contrast switches the intervention distribution of the target mediator while assigning the mediators before it to their distributions under $A = 0$ and those after it to their distributions under $A = 1$. Reversing the ordering changes the intervention distribution assigned to the nontarget mediator and may therefore change the mediator-specific estimand. This example also clarifies why the ordering should be prespecified scientifically and why alternative plausible orderings can be examined as a sensitivity analysis.

B Additional Synthetic Experiment with Dependent Mediators

B.1 Dependent-Mediator Experiment

To assess joint mediator learning beyond the primary synthetic experiment, we consider a multivariate setting with $S = 5$ dependent mediators. The mediator vector is generated as

$$\boldsymbol{M} = 0.5\,\mathbf{1}_5 + \boldsymbol{b}_A A + \boldsymbol{b}_Z Z + \boldsymbol{\varepsilon}_M, \qquad \boldsymbol{\varepsilon}_M \sim \mathcal{N}(\mathbf{0}_5, \Sigma),$$

where $\Sigma \in \mathbb{R}^{5\times 5}$ has entries $\Sigma_{ij} = 0.6^{|i-j|}$, $\boldsymbol{b}_A = (1.0, 0.8, 0.6, 0.4, 0.2)$, and $\boldsymbol{b}_Z = (0.3, 0.3, 0.2, 0.2, 0.1)$. The outcome is generated as

$$Y = 1 + 0.6A + 0.2\,\mathbf{1}_5^\top \boldsymbol{M} + \sin(M_1 M_2) + 0.2Z + \varepsilon_Y, \qquad \varepsilon_Y \sim N(0,1).$$

Treatment, covariate, sample-size, and replication settings are the same as in the primary synthetic experiment in the main text. This setting combines dependent mediators with a nonlinear mediator interaction in the outcome model.

Table 1: Dependent-mediators experiment. Entries are RMSEs for mean-based and ED-based interventional mediation effect estimates over 100 replications.

Metric	IDE	IIE	IPSE_1	IPSE_2	IPSE_3	IPSE_4	IPSE_5
Mean RMSE	0.038	0.020	0.023	0.015	0.010	0.007	0.005
ED RMSE	0.013	0.005	0.003	0.002	0.001	0.001	0.000

B.2 Joint Mediator Modeling Ablation

We also compare the proposed joint mediator generator with a variant that models each mediator marginally and combines the generated draws during interventional reconstruction. The two specifications have similar IPSE errors, but separate mediator modeling yields larger ED RMSEs for the IIE and ITE.

Table 2: Joint mediator modeling ablation in the dependent-mediators setting. Entries are ED RMSEs over 100 replications.

Variant	ITE	IDE	IIE	IPSE_1	IPSE_2	IPSE_3	IPSE_4	IPSE_5
Joint	**0.020**	**0.013**	**0.005**	**0.003**	**0.002**	**0.001**	**0.001**	0.000
Separate	0.043	0.015	0.014	0.004	0.002	0.001	0.001	**0.000**

C Semi-Synthetic IHDP Data-Generating Mechanism

We retain the observed IHDP treatment indicator $A_i \in \{0,1\}$ and use the first ten baseline covariates as $\boldsymbol{Z}_i = (Z_{i1}, \dots, Z_{i10})$.

The mediators are generated as

$$\begin{aligned} M_{i1} &= 0.10 + 0.55A_i + 0.20Z_{i1} - 0.15Z_{i2} + 1.10\,\varepsilon_{i1}, \\ M_{i2} &= -0.15 + 0.60A_i + 0.10Z_{i4} + 1.10\,\varepsilon_{i2}, \\ M_{i3} &= -0.70 + 0.65A_i + 0.08Z_{i5} + 1.10\,\varepsilon_{i3}, \end{aligned}$$

where

$$\varepsilon_{i1}, \varepsilon_{i2}, \varepsilon_{i3} \overset{\text{i.i.d.}}{\sim} N(0,1).$$

The outcome is generated as

$$Y_i = \mu_i + \sigma_i R_i,$$

where

$$\begin{aligned}
\mu_i &= 1.00 + 0.98M_{i1} + 0.08Z_{i1} - 0.06Z_{i2},\\
\log\sigma_i &= 0.55M_{i2},\\
R_i &= U_i + 10\ \mathrm{expit}\{5(M_{i3} + 0.05)\}\, g_i, \qquad U_i \sim N(0,1),
\end{aligned}$$

and

$$g_i = \mathrm{expit}\{5(Z_{i5} - \tau_G)\} - \frac{1}{n}\sum_{\ell=1}^{n} \mathrm{expit}\{5(Z_{\ell 5} - \tau_G)\}.$$

Here τ_G is the empirical 0.88 quantile of $\{Z_{i5} : i = 1, \dots, n\}$, and $\mathrm{expit}(u) = \{1 + \exp(-u)\}^{-1}$. The centered gate g_i is positive mainly for subjects with large Z_{i5} and slightly negative for most others. Thus, larger values of M_{i3} amplify positive residual values mainly in a small high-Z_{i5} subgroup, producing a heavier right tail and greater tail asymmetry.

D Implementation Details

Table 3 summarizes the generator architectures used for the main experiments. Unless otherwise stated, all generators are trained with the ES objective using Adam with learning rate 5×10^{-4}, a 20% validation split, and early stopping. The ES-based loss uses $L_{\mathrm{ES}} = 20$ generator draws.

Table 3: Generator architectures used in the main experiments. The notation $L \times H$ denotes L hidden layers with hidden width H.

Experiment	Noise dimension		Network architecture	
	ε_M	ε_Y	Mediator model	Outcome model
Primary synthetic	4	8	5×64	5×64
Dependent mediators	4	8	5×64	5×64
IHDP	4	32	5×64	7×128
NHANES	4	8	3×64	3×64

E Identification Proofs

E.1 Proof of Theorem 1

We first prove the formula for $R^{z}_{a,a'}$. Conditional on $\boldsymbol{Z} = \boldsymbol{z}$, the stochastic mediator draw $\tilde{\boldsymbol{M}}_{a'}$ is drawn from the interventional mediator distribution induced by treatment level a'. Therefore,

$$\mathbb{P}(Y_{a\tilde{\boldsymbol{M}}_{a'}} \in B \mid \boldsymbol{Z} = \boldsymbol{z}) = \int \mathbb{P}(Y_{a\boldsymbol{m}} \in B \mid \boldsymbol{Z} = \boldsymbol{z})\, P_{\tilde{\boldsymbol{M}}_{a'}|\boldsymbol{Z}=\boldsymbol{z}}(d\boldsymbol{m}).$$

By the definition of the interventional mediator draw, treatment–mediator ignorability, and consistency,

$$P_{\tilde{\boldsymbol{M}}_{a'}|\boldsymbol{Z}=\boldsymbol{z}} = P_{\boldsymbol{M}|A=a',\boldsymbol{Z}=\boldsymbol{z}} = G^{z}_{a'}.$$

Moreover, by sequential ignorability and consistency,

$$\begin{aligned}
\mathbb{P}(Y_{a\boldsymbol{m}} \in B \mid \boldsymbol{Z} = \boldsymbol{z}) &= \mathbb{P}(Y_{a\boldsymbol{m}} \in B \mid A = a, \boldsymbol{M} = \boldsymbol{m}, \boldsymbol{Z} = \boldsymbol{z})\\
&= \mathbb{P}(Y \in B \mid A = a, \boldsymbol{M} = \boldsymbol{m}, \boldsymbol{Z} = \boldsymbol{z})\\
&= K^{z}_{a}(B \mid \boldsymbol{m}).
\end{aligned}$$

Combining the two displays gives

$$\mathbb{P}(Y_{a\tilde{\boldsymbol{M}}_{a'}} \in B \mid \boldsymbol{Z} = \boldsymbol{z}) = \int K_a^z(B \mid \boldsymbol{m})\, G_{a'}^z(d\boldsymbol{m}) = R_{a,a'}^z(B).$$

Integrating both sides with respect to $P_{\boldsymbol{Z}}(d\boldsymbol{z})$ yields the marginal distribution $R_{a,a'}$.

E.2 Expanded IPSE Identification Formula

For $r \in \{0, 1\}$, the IPSE-related mediator intervention is

$$Q_{s,r}^z = G_0^{z,<s} \otimes G_r^{z,s} \otimes G_1^{z,>s}.$$

Hence, for any Borel set $B \subseteq \mathbb{R}$,

$$\begin{aligned}
&\mathbb{P}\Big(Y_{1\tilde{\boldsymbol{M}}_0^{(<s)}\tilde{M}_{sr}\tilde{\boldsymbol{M}}_1^{(>s)}} \in B \mid \boldsymbol{Z} = \boldsymbol{z}\Big) \\
&\quad = \int K_1^z\Big(B \mid \boldsymbol{m}^{(<s)}, m_s, \boldsymbol{m}^{(>s)}\Big) \\
&\qquad \times G_0^{z,<s}(d\boldsymbol{m}^{(<s)}) G_r^{z,s}(dm_s) G_1^{z,>s}(d\boldsymbol{m}^{(>s)}).
\end{aligned}$$

This is the expanded form of

$$R_{s,r}^z(B) = \int K_1^z(B \mid \boldsymbol{m})\, Q_{s,r}^z(d\boldsymbol{m}).$$

The marginal distribution $R_{s,r}$ is obtained by integrating $R_{s,r}^z$ over $P_{\boldsymbol{Z}}(d\boldsymbol{z})$. Therefore,

$$\mathrm{IPSE}_s^\Psi = \Psi(R_{s,1}, R_{s,0}).$$

The proof is the same as the proof for $R_{a,a'}^z$, replacing the joint mediator intervention distribution $G_{a'}^z$ with the product intervention distribution $Q_{s,r}^z$ and setting the outcome treatment level to $A = 1$.

F Proofs for the ED-Based Error Analysis

F.1 Proof of Theorem 2

For brevity, write

$$\widehat{K}_{\boldsymbol{m}} = \widehat{K}_a^z(\cdot \mid \boldsymbol{m}), \qquad K_{\boldsymbol{m}} = K_a^z(\cdot \mid \boldsymbol{m}).$$

Define the intermediate distribution

$$R^{z,\circ} = \int \widehat{K}_a^z(\cdot \mid \boldsymbol{m})\, H^z(d\boldsymbol{m}).$$

By the triangle inequality in $\mathcal{H}_Y$,

$$\mathsf{d}_Y(\widehat{R}^z, R^z) \leq \mathsf{d}_Y(\widehat{R}^z, R^{z,\circ}) + \mathsf{d}_Y(R^{z,\circ}, R^z).$$

For the mediator-stage term, linearity of kernel mean embeddings gives

$$\mu_Y(\widehat{R}^z) - \mu_Y(R^{z,\circ}) = \int \widehat{\phi}_{a,z}(\boldsymbol{m})\{\widehat{H}^z - H^z\}(d\boldsymbol{m}).$$

For any $v \in \mathcal{H}_Y$, define the scalar projection

$$g_v(\boldsymbol{m}) = \langle v, \widehat{\phi}_{a,z}(\boldsymbol{m}) \rangle_{\mathcal{H}_Y}.$$

The embedding regularity condition in Theorem 2 implies that $g_v \in \mathcal{H}_M$ and

$$\|g_v\|_{\mathcal{H}_M} \le L_\phi \|v\|_{\mathcal{H}_Y}.$$

Therefore, for any $v \in \mathcal{H}_Y$ with $\|v\|_{\mathcal{H}_Y} \le 1$, the reproducing property of kernel mean embeddings and the Cauchy–Schwarz inequality yield

$$\begin{aligned} \left| \left\langle v, \mu_Y(\widehat{R}^z) - \mu_Y(R^{z,\circ}) \right\rangle_{\mathcal{H}_Y} \right| &= \left| \int g_v(\boldsymbol{m}) \{\widehat{H}^z - H^z\}(d\boldsymbol{m}) \right| \\ &= \left| \left\langle g_v, \mu_M(\widehat{H}^z) - \mu_M(H^z) \right\rangle_{\mathcal{H}_M} \right| \\ &\le L_\phi \, \mathsf{d}_M(\widehat{H}^z, H^z). \end{aligned}$$

Taking the supremum over all v in the unit ball of $\mathcal{H}_Y$ gives

$$\mathsf{d}_Y(\widehat{R}^z, R^{z,\circ}) \le L_\phi \, \mathsf{d}_M(\widehat{H}^z, H^z).$$

For the outcome-stage term,

$$\mu_Y(R^{z,\circ}) - \mu_Y(R^z) = \int \{\mu_Y(\widehat{K_{\boldsymbol{m}}}) - \mu_Y(K_{\boldsymbol{m}})\} \, H^z(d\boldsymbol{m}).$$

Jensen's inequality for Bochner integrals yields

$$\begin{aligned} \mathsf{d}_Y(R^{z,\circ}, R^z) &\le \int \mathsf{d}_Y(\widehat{K_{\boldsymbol{m}}}, K_{\boldsymbol{m}}) \, H^z(d\boldsymbol{m}) \\ &\le \left\{ \int \mathsf{d}_Y^2(\widehat{K_{\boldsymbol{m}}}, K_{\boldsymbol{m}}) \, H^z(d\boldsymbol{m}) \right\}^{1/2}. \end{aligned}$$

Combining the mediator-stage and outcome-stage bounds proves

$$\mathsf{d}_Y(\widehat{R}^z, R^z) \le L_\phi \, \mathsf{d}_M(\widehat{H}^z, H^z) + \left\{ \int \mathsf{d}_Y^2(\widehat{K_{\boldsymbol{m}}}, K_{\boldsymbol{m}}) \, H^z(d\boldsymbol{m}) \right\}^{1/2}.$$

Since

$$\mathrm{ED}_Y(P, Q) = 2\mathsf{d}_Y^2(P, Q), \qquad \mathrm{ED}_M(P, Q) = 2\mathsf{d}_M^2(P, Q),$$

and $(x + y)^2 \le 2x^2 + 2y^2$, we obtain

$$\begin{aligned} \mathrm{ED}_Y(\widehat{R}^z, R^z) &\le 4L_\phi^2 \mathsf{d}_M^2(\widehat{H}^z, H^z) + 4 \int \mathsf{d}_Y^2(\widehat{K_{\boldsymbol{m}}}, K_{\boldsymbol{m}}) H^z(d\boldsymbol{m}) \\ &= 2L_\phi^2 \mathrm{ED}_M(\widehat{H}^z, H^z) + 2 \int \mathrm{ED}_Y(\widehat{K_{\boldsymbol{m}}}, K_{\boldsymbol{m}}) H^z(d\boldsymbol{m}). \end{aligned}$$

This proves the theorem.

G Plug-in Error Bounds for Effect Functionals

Let P and Q denote any pair of target interventional outcome distributions whose effect is being evaluated, and let $\widehat{P}$ and $\widehat{Q}$ denote their reconstructed counterparts. Define

$$\Delta = \mathsf{d}_Y(\widehat{P}, P) + \mathsf{d}_Y(\widehat{Q}, Q).$$

We first consider a contrast functional

$$\Psi_T(P, Q) = T(P) - T(Q),$$

where T is a scalar summary of an outcome distribution. A sufficient uniform stability condition is that, on the relevant class of distributions, T is Lipschitz continuous with respect to d_Y, that is,

$$|T(P') - T(Q')| \leq L_T\, \mathsf{d}_Y(P', Q')$$

for all distributions P' and Q' in this class and for some $L_T < \infty$. Then

$$\left|\Psi_T(\widehat{P}, \widehat{Q}) - \Psi_T(P, Q)\right| \leq \left|T(\widehat{P}) - T(P)\right| + \left|T(\widehat{Q}) - T(Q)\right| \leq L_T \Delta.$$

Thus, whenever the chosen scalar summary is stable with respect to d_Y, the plug-in error of its contrast is controlled by the reconstruction errors of the two interventional outcome distributions.

This Lipschitz condition is sufficient rather than necessary and depends on the distribution class. For example, for univariate outcomes supported on a common bounded interval $[a, b]$, the energy metric satisfies

$$\mathsf{d}_Y^2(P, Q) = \int_a^b \{F_P(y) - F_Q(y)\}^2\, dy,$$

and the mean functional obeys

$$|\mathbb{E}_P(Y) - \mathbb{E}_Q(Y)| \leq \sqrt{b-a}\, \mathsf{d}_Y(P, Q).$$

On unbounded supports, the mean need not be uniformly Lipschitz without additional tail restrictions. Quantiles and exceedance probabilities also need not be Lipschitz with respect to d_Y on unrestricted distribution classes; their error control requires additional functional-specific regularity, such as local density and continuity conditions. The contrast bound above therefore does not automatically cover every scalar summary considered in the main text.

We next consider the ED effect

$$\Psi_{\mathrm{ED}}(P, Q) = ED_Y(P, Q) = 2\mathsf{d}_Y^2(P, Q).$$

Since d_Y is a metric, the reverse triangle inequality gives

$$\left|\mathsf{d}_Y(\widehat{P}, \widehat{Q}) - \mathsf{d}_Y(P, Q)\right| \leq \Delta.$$

Therefore,

$$\begin{aligned}
\left|\Psi_{\mathrm{ED}}(\widehat{P}, \widehat{Q}) - \Psi_{\mathrm{ED}}(P, Q)\right| &= 2\left|\mathsf{d}_Y^2(\widehat{P}, \widehat{Q}) - \mathsf{d}_Y^2(P, Q)\right| \\
&\leq 2\Delta\left\{\mathsf{d}_Y(\widehat{P}, \widehat{Q}) + \mathsf{d}_Y(P, Q)\right\} \\
&\leq 2\Delta\left\{2\mathsf{d}_Y(P, Q) + \Delta\right\}.
\end{aligned}$$

In particular, on any class of distribution pairs satisfying $\mathsf{d}_Y(P, Q) \leq B$,

$$\left|\Psi_{\mathrm{ED}}(\widehat{P}, \widehat{Q}) - \Psi_{\mathrm{ED}}(P, Q)\right| \leq 4B\Delta + 2\Delta^2.$$

Thus, the plug-in error of the ED effect is also controlled by the reconstruction errors of the two interventional outcome distributions.